

Microservices Architecture Overview

Microservices architecture decomposes applications into small, independent services. Each service communicates via lightweight REST APIs or gRPC protocols.

Key Benefits

1. Fault Isolation: A crash in one module does not bring down the whole application.
2. Scalability: Individual services can scale horizontally based on demand.
3. Deployment Speed: Independent CI/CD pipelines allow rapid delivery.

Common Challenges

Data Consistency: Distributed transactions require event-driven architecture such as the Saga pattern.

Service Discovery: Dynamic IP management requires tools like Consul or Kubernetes DNS.